Task 2: Huntington's Disease and the Occipital Lobe

Intro to Computational Medicine: Imaging · Fall 2026 Final Project · **Group 1**

Dataset: HD_ICM_random200.xlsx · ROIs (left and right of each, 10 volumes): cuneus (Cu), lingual gyrus (LG), inferior occipital gyrus (IOG), middle occipital gyrus (MOG), superior occipital gyrus (SOG)

Task 2 asks us to *discuss how the brain region may be involved in the assigned disease and summarize relevant existing literature* (grading: involvement 2 pts, literature 2 pts). The matching slides are in Task2_Slides.pptx. Every claim below is tied to a source in the reference list.

1. Summary

HD is best known as a disease of the striatum, but the mutant protein is made by neurons throughout the brain and the cerebral cortex degenerates too. Among cortical regions, the occipital lobe is one of the earliest and fastest to thin. In TRACK-HD (the study our data come from), occipital thinning appears even before symptoms and predicts worse visual-cognition scores. We therefore expect HD subjects to have smaller occipital volumes than controls, and we test this in Task 4.

2. The disease in brief

- **Cause.** An expanded CAG trinucleotide repeat in the huntingtin gene (*HTT*, chromosome 4p16.3), inherited in an autosomal dominant pattern [HDCRG 1993]. 36–39 repeats give reduced penetrance; 40 or more give full penetrance [Tabrizi 2020].
- **Hallmark pathology.** Loss of GABAergic medium spiny neurons in the striatum (caudate and putamen) [Tabrizi 2020].
- **Symptoms.** A triad of motor (chorea), cognitive and psychiatric symptoms, usually starting between ages 30 and 50 [HDSA].
- **Silent phase.** Measurable brain and cognitive changes begin roughly 10–20 years before clinical diagnosis [Paulsen 2008].

3. How the occipital lobe could be involved

1. **The cortex degenerates too, not just the striatum.** Huntingtin is expressed widely [HDCRG 1993]. Post-mortem stereology finds loss of total neurons and pyramidal cells across frontal, parietal, temporal and occipital cortex [Nana 2014].
2. **Thinning moves from back to front.** Cortical thinning starts early and appears to spread from posterior to anterior regions as the disease progresses [Rosas 2002]. In stage I–II patients, sensorimotor, parietal and occipital cortex are already thin, while anterior frontal and temporal regions are relatively spared [Rosas 2008].
3. **The visual cortex is wired into the striatum.** Visual cortex projects to the tail of the caudate through a visual corticostriatal loop [Seger 2013]. Damage at either end of a corticostriatal loop is one proposed route by which striatal and cortical degeneration spread to each other.

4. **Clinical consequences.** People with HD show reduced integrity of the visual pathway and deficits in color and motion perception [Dhalla 2019]. They also show impaired visual search and mental rotation, which track posterior brain structure [Labuschagne 2016]. Visual deficits matter for daily function and for cognitive tests used in HD trials [Coppen 2018, *Behav Brain Res*].

4. Literature evidence

Year	Study	Sample and method	Occipital finding
2002	Rosas et al., *Neurology*	11 HD vs 13 controls; MRI cortical thickness	Thinning occurs early and appears to proceed from posterior to anterior; sensorimotor cortex most affected
2008	Rosas et al., *Brain*	33 HD across stages vs 22 controls; MRI cortical thickness	Stage I–II: thinning over sensorimotor, parietal and occipital cortex; anterior frontal and temporal relatively spared. Regional thinning tracks clinical heterogeneity
2014	Nana et al., *J Huntingtons Dis*	14 HD vs 15 controls; post-mortem stereology, 6 cortical regions	Neuron and pyramidal-cell loss in every region studied **except primary visual cortex**
2015	Rüb et al., *Brain Pathol*	7 HD vs 7 controls; stereology of BA17 (primary visual cortex)	**32% fewer neurons** in BA17 (≈71 vs ≈104 million), mainly in layers III, IV and VI
2015	Johnson et al., *Neuropsychologia*	TRACK-HD: 97 controls, 109 premanifest, 69 early HD; FreeSurfer thickness of 4 occipital regions	Occipital thickness reduced in **both premanifest and early HD**. Thinner cortex linked to worse performance on 5 of 6 visual tasks, strongest for **lateral occipital and lingual**; **cuneus showed no association**; finger tapping (non-visual) unrelated
2016	Labuschagne et al., *J Int Neuropsychol Soc*	TRACK-HD: 119 premanifest, 104 early HD, 110 controls	Visual search and mental rotation track parietal and occipital volume and thickness (precuneus, cuneus, calcarine, lingual) and posterior fusiform. Visual search declined over 12 months in early HD
2018	Coppen et al., *Hum Brain Mapp*	18 controls, 21 premanifest, 20 manifest; VBM, thickness, resting-state fMRI	Manifest HD: atrophy of associative visual cortex and reduced connectivity of lingual gyrus, occipital pole and occipital fusiform with the medial visual network. **Premanifest: no differences**
2019	Dhalla et al., *J Huntingtons Dis*	Systematic review	Retinal thinning, reduced VEP amplitude, visual-cortex cell loss and thinning; color and motion perception deficits
2021	Johnson et al., *Biol Psychiatry*	49 HD gene carriers vs 49 controls, 3–7 scans each over 10 years	Widespread cortical differences, but **occipital and parietal regions show the greatest rate of atrophy**

Where the literature disagrees (useful for Task 4)

- **Primary visual cortex (V1).** Nana 2014 found V1 spared, while Rüb 2015 found 32% neuron loss in V1. Associative visual cortex (the lateral occipital and lingual regions) is affected more consistently.
- **Premanifest stage.** Johnson 2015 (n = 109 premanifest) found occipital thinning before symptoms, while Coppen 2018 (n = 21 premanifest) did not. The larger study is more likely to detect small early effects.
- **Our dataset.** The spreadsheet labels subjects only as HD or NORMAL. TRACK-HD and PREDICT-HD both enrolled premanifest gene carriers [Tabrizi 2009; Paulsen 2008]. Our HD group is therefore

probably a mix of premanifest and early-stage people. If so, our effect sizes should be smaller than in studies of manifest HD only.

5. Matching literature labels to our ROIs

Most of the imaging studies above use FreeSurfer's Desikan–Killiany labels. Our volumes come from MRICloud's JHU multi-atlas labels, which divide the occipital lobe differently:

Literature (FreeSurfer)	Our MRICloud ROIs
Lateral occipital	IOG, MOG and part of SOG
Lingual	LG
Cuneus	Cu
Pericalcarine (V1, calcarine banks)	No separate label: upper bank in Cu, lower bank in LG

So when we compare our Task 4 results with Johnson 2015, "lateral occipital" findings should be compared with our IOG and MOG results.

6. Hypothesis carried into Task 4

- **H1.** HD subjects have smaller volumes than controls in all 10 occipital ROIs, on both sides.
- **H2.** The lingual and lateral occipital gyri (LG, IOG, MOG) are the regions most tied to symptoms, because they have the strongest links to visual-cognition deficits [Johnson 2015; Coppen 2018].

H2 is about clinical relevance, not a prediction that these ROIs show the largest volume differences. The literature does not rank the five gyri by atrophy size.

7. Speaker script (about 80 seconds)

Slide "Beyond the striatum" (about 25 s)

Huntington's disease is caused by an expanded CAG repeat in the huntingtin gene and is inherited in an autosomal dominant way. Its hallmark is the loss of medium spiny neurons in the striatum. But mutant huntingtin is made by neurons throughout the brain, so the cortex degenerates too, and cortical thinning moves from the back of the brain forward. That includes the visual cortex, which loops into the tail of the caudate. Patients develop problems with color and motion perception, visual search and mental rotation.

Slide "Studies place the occipital lobe early in HD" (about 30 s)

The literature consistently puts the occipital lobe early in the disease. Rosas showed that cortical thinning moves from posterior to anterior, with occipital thinning already present in stage one and two patients. Post-mortem counts confirm neuron loss across the cortex, though studies disagree about the primary visual cortex itself. In TRACK-HD, the study our data come from, Johnson found occipital thinning even before symptoms, and it predicted worse visual-test scores. Over ten years, the occipital and parietal cortex atrophied fastest of any cortical region.

Slide “What we expect in our 10 occipital ROIs” (about 25 s)

For our ten structures, the lingual gyrus and the lateral occipital gyri, inferior and middle, have the strongest links to visual deficits in HD. The cuneus thins but has not been tied to test scores. So we hypothesize that HD subjects have smaller volumes in all ten occipital ROIs, on both sides, which we test next. One caveat: most studies use FreeSurfer labels, so their “lateral occipital” roughly equals our IOG and MOG, and primary visual cortex is split between our cuneus and lingual gyrus.

8. References

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Further reading (not cited on the slides): Wolf RC, et al. Visual system integrity and cognition in early Huntington's disease. *Eur J Neurosci.* 2014;40:2417–2426. doi:10.1111/ejn.12575. This study found occipitotemporal atrophy, and its reduced fusiform activity tracked lower SDMT scores.